

# Euclid's Elements — Book I

## Proposition 26

Formal Reconstruction — Technical Documentation

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*If two triangles have two angles equal to two angles respectively, and one side equal to one side, namely, either the side adjoining the equal angles, or that subtending one of the equal angles, then they will also have the remaining sides equal to the remaining sides and the remaining angle equal to the remaining angle.*

## 1 Introduction

Euclid's Proposition I.26 contains two triangle-congruence criteria.

For two nondegenerate triangles  $ABC$  and  $DEF$ , assume the two corresponding angles

$$\angle ABC \cong \angle DEF, \quad \angle BCA \cong \angle EFD.$$

Euclid then considers two possibilities for the given corresponding side.

1. In the ASA case,

$$BC \cong EF.$$

2. In the SAA case,

$$AB \cong DE.$$

In either case the conclusion is full triangle congruence:

$$AB \cong DE, \quad BC \cong EF, \quad AC \cong DF,$$

together with

$$\angle BAC \cong \angle EDF.$$

The two parts have very different formal behavior.

The ASA part is almost entirely absorbed by the existing Hilbert congruence theory. The SAA part is not. Its proof still requires the essential contradiction argument of Euclid: a hypothetical strict inequality of the corresponding sides is converted into an auxiliary SAS configuration and then contradicted by Proposition I.16.

Thus Proposition I.26 provides a useful boundary case. One half is already available as interface-level geometry; the other still exposes genuine Book I proof structure.

## 2 Classical Configurations

Wilkins emphasizes that Proposition I.26 establishes both the ASA and the SAA congruence rules. The same basic pair of triangles is used for both arguments, but the auxiliary point lies on a different side in the two cases.

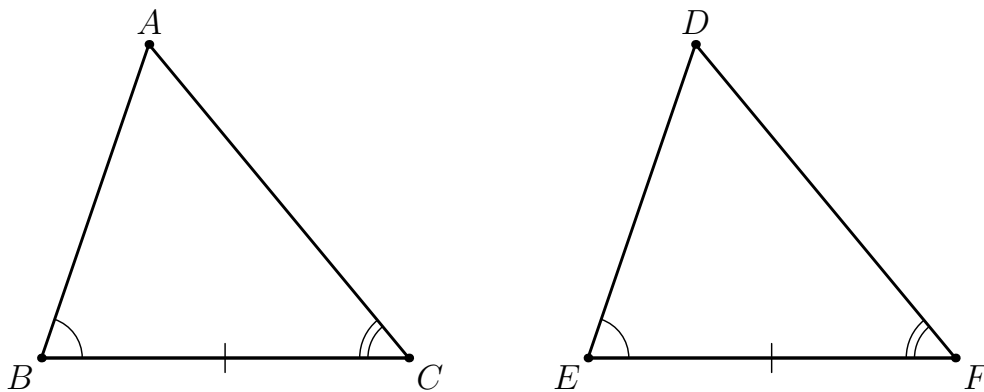


Figure 1: The common input configuration for Proposition I.26. The two pairs of corresponding angles are marked by one and two arcs respectively, and in the ASA case the corresponding side  $BC \cong EF$  is marked by a single tick.

### 2.1 ASA auxiliary configuration

Assume first

$$BC \cong EF.$$

Euclid supposes, for contradiction, that the corresponding sides  $AB$  and  $DE$  are unequal. If  $AB$  is the greater side, choose a point  $G$  between  $A$  and  $B$  such that

$$GB \cong DE,$$

and join  $G$  to  $C$ .

SAS gives

$$\angle BCG \cong \angle EFD.$$

Together with the original hypothesis

$$\angle EFD \cong \angle BCA,$$

this would imply

$$\angle BCG \cong \angle BCA.$$

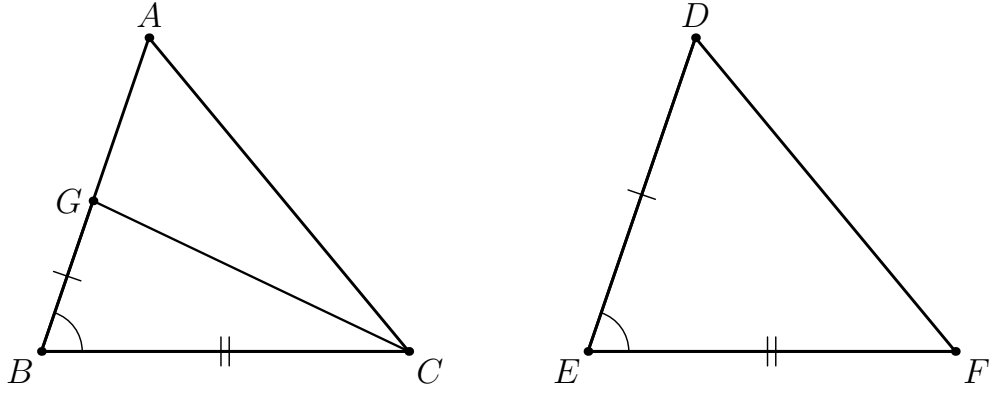


Figure 2: Euclid's ASA reductio configuration. The point  $G$  lies between  $A$  and  $B$ . The diagram marks exactly the local SAS data  $GB \cong DE$ ,  $BC \cong EF$ , and  $\angle GBC \cong \angle DEF$ .

But  $G$  lies inside the side  $AB$ , so  $\angle BCG$  is a proper part of  $\angle BCA$ . This contradiction forces

$$AB \cong DE.$$

A final SAS application to the original triangles gives the remaining side and angle congruences.

## 2.2 SAA auxiliary configuration

Now assume instead

$$AB \cong DE.$$

Suppose, for contradiction, that  $EF < BC$ . Then there is a point  $H$  between  $B$  and  $C$  such that

$$BH \cong EF.$$

Join  $A$  to  $H$ .

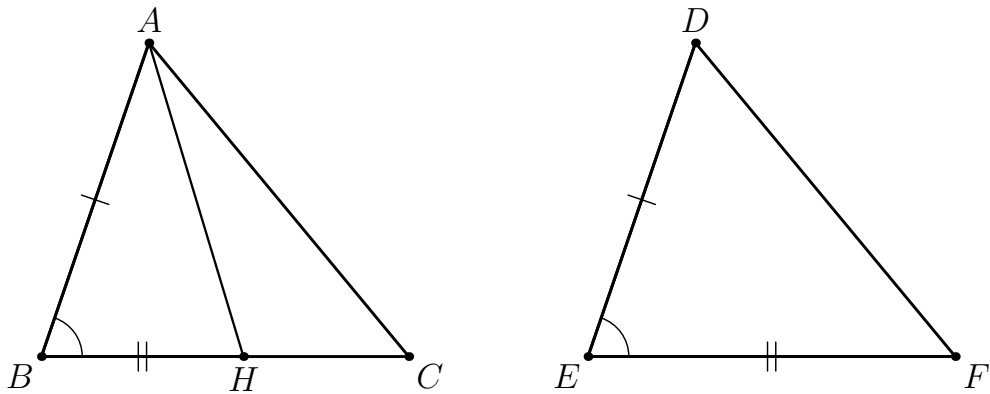


Figure 3: Euclid's SAA reductio configuration. The point  $H$  lies between  $B$  and  $C$ . The diagram marks exactly the local SAS data  $AB \cong DE$ ,  $BH \cong EF$ , and  $\angle ABH \cong \angle DEF$ . Proposition I.16 is then applied to  $\triangle AHC$ .

Since  $B, H, C$  are collinear on the same ray from  $B$ ,

$$\angle ABH \cong \angle ABC.$$

Hence SAS on  $\triangle BAH$  and  $\triangle EDF$  gives

$$\angle BHA \cong \angle EFD.$$

Using the original angle hypothesis,

$$\angle EFD \cong \angle BCA,$$

we obtain

$$\angle BHA \cong \angle BCA.$$

On the other hand,  $C - H - B$ , so  $\angle BHA$  is an exterior angle of triangle  $AHC$ . Proposition I.16 gives the strict inequality

$$\angle BCA < \angle BHA,$$

contradicting their congruence.

Thus

$$\neg(EF < BC).$$

Interchanging the two triangles and repeating the same argument gives

$$\neg(BC < EF).$$

Segment trichotomy therefore yields

$$BC \cong EF.$$

The SAA case has now been reduced to the ASA case.

### 3 Diagrammatic Audit

Three figures are retained because they have three genuinely different jobs.

Figure 1 records the common input configuration. Figure 2 isolates the auxiliary SAS configuration used in the ASA reductio. Figure 3 isolates the corresponding SAS configuration used in the SAA reductio.

The last two diagrams deliberately do not mark every hypothesis of Proposition I.26. They mark only the congruences used in the local SAS step. This follows the logic of Wilkins's presentation and avoids overloading a single figure with relations that belong to different stages of the proof.

The auxiliary points are constructed geometrically:  $G$  lies genuinely on the segment  $AB$ , and  $H$  lies genuinely on the segment  $BC$ . Thus the incidence information used by the proof is represented directly rather than being approximated by hand-chosen coordinates.

These are not successive snapshots of one construction. They represent the two different reductio mechanisms contained in Proposition I.26.

### 4 Hilbert Reconstruction

The two halves of Proposition I.26 behave differently over the present Hilbert interface.

## 4.1 ASA

The theorem `hilbert_asa_sides` already contains the central mathematical content of the ASA case.

The given side in Euclid’s formulation is

$$BC \cong EF.$$

The theorem is applied after cyclically viewing the triangles as

$$\triangle BCA \quad \text{and} \quad \triangle EFD.$$

After the required angle and endpoint reorientations it gives

$$BA \cong ED, \quad CA \cong FD.$$

These are transported to

$$AB \cong DE, \quad AC \cong DF.$$

At that point all three pairs of sides are congruent. The existing `HilbertSSS` theorem supplies the remaining corresponding angle:

$$\angle BAC \cong \angle EDF.$$

Thus the formal ASA proof does not reconstruct Euclid’s reductio.

## 4.2 SAA

No permutation of `hilbert_asa_sides` directly solves the SAA problem. With the side  $AB \cong DE$  fixed, ASA would require the angles at  $A$  and  $B$ , while the hypotheses supply the angles at  $B$  and  $C$ .

The formal reconstruction therefore isolates Euclid’s essential contradiction argument in the helper

```
euclid_proposition_26_saa_not_less
```

The helper proves

$$\neg(EF < BC).$$

Its architecture is

```
EF < BC
|
witness H with B-H-C and BH ~= EF
|
same-ray transport:
angle ABH ~= angle ABC
|
SAS on BAH and EDF
```

```

      |
angle BHA ~= angle EFD ~= angle BCA
      |
I.16 on triangle ACH
      |
angle BCA < angle BHA
      |
contradiction

```

The same helper is then called again with the two triangles interchanged, producing

$$\neg(BC < EF).$$

Book Zero segment trichotomy yields

$$BC \cong EF,$$

and the public SAA theorem finishes by calling the already proved ASA case.

## 5 Proof Architecture of the Reconstruction

The two formal paths can be summarized separately.

I.26 ASA

-----

```

two angle congruences
+ BC ~= EF
      |
      v
hilbert_asa_sides
on BCA and EFD
      |
      v
AB ~= DE
AC ~= DF
      |
      v
HilbertSSS
      |
      v
angle BAC ~= angle EDF

```

I.26 SAA

-----

```

two angle congruences
+ AB ~= DE
      |
      +-----+
      |                               |

```

|                                                                                                                                                                                                                          |                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\downarrow$<br>not (EF < BC)<br>SAS + I.16<br><br> <br>+-----+-----+<br> <br>$\downarrow$<br>bookZero_31_trichotomy1<br> <br>$\downarrow$<br>BC $\approx$ EF<br> <br>$\downarrow$<br>I.26 ASA case<br> <br>$\downarrow$ | $\downarrow$<br>not (BC < EF)<br>same helper,<br>triangles swapped<br> <br>+-----+-----+<br> <br>$\downarrow$<br>bookZero_31_trichotomy1<br> <br>$\downarrow$<br>BC $\approx$ EF<br> <br>$\downarrow$<br>I.26 ASA case<br> <br>$\downarrow$ |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

AC  $\approx$  DF  
angle BAC  $\approx$  angle EDF

The contrast with Euclid is precise.

For ASA, the Hilbert interface has already absorbed the reductio into a general theorem.

For SAA, Proposition I.16 remains a genuine formal dependency.

## 6 Lean Formalization

The complete implementation is available in the repository. Only the decisive interface steps are reproduced here.

For ASA, the first substantial call is

```
have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    E F D
    hBCA
    hEFD
    hBC
    hAngleB'
    hAngleC
```

After endpoint reversal this gives the two missing sides. The final angle is then obtained from

```
have hSSS :=
  HilbertSSS
    Geo
    A B C
    D E F
    hABC
```

```

hAB
hBC
hAC

```

For SAA, the public proof first excludes both strict segment inequalities:

```

have hNotEF_BC :=
  euclid_proposition_26_saa_not_less
  Geo A B C D E F
  hABC hDEF hAngleB hAngleC hAB

```

The same helper is applied with the two triangles interchanged. Segment trichotomy then produces the missing side congruence, and the theorem reduces to the already established ASA case.

This organization keeps the long exterior-angle argument inside one named helper rather than duplicating it in the public theorem.

## 7 Relationship with Earlier Propositions

### 7.1 Proposition I.4

Euclid uses SAS in both halves of I.26.

In the ASA proof, SAS is used after cutting off a segment congruent to the corresponding side of the second triangle.

In the SAA proof, SAS is applied to the auxiliary triangle  $BAH$  and  $\triangle EDF$ .

The formal SAA helper preserves this use of SAS directly.

### 7.2 Proposition I.16

Proposition I.16 is the decisive difference between the ASA and SAA arguments.

The SAA helper requires the exterior-angle inequality

$$\angle BCA < \angle BHA.$$

This strict inequality contradicts the angle congruence obtained from SAS. Without I.16 the helper would not close.

### 7.3 Segment trichotomy

The formal SAA proof separates the two possible strict segment inequalities:

$$EF < BC \quad \text{and} \quad BC < EF.$$

After both are excluded, `bookZero_31_trichotomy1` returns the remaining possibility

$$BC \cong EF.$$



Thus Book Zero supplies the final order-theoretic bridge from the two contradiction arguments to the ASA case.

## 8 Hilbert and Book Zero Components Used

### 8.1 `hilbert_asa_sides`

This is the main interface theorem for the ASA half. It converts a common corresponding side together with the two adjacent angle congruences into the remaining two side congruences.

Its availability explains why the formal ASA proof is much shorter than Euclid’s proof.

### 8.2 `HilbertSSS`

Once all three corresponding side congruences have been recovered, SSS provides the remaining angle congruence required by the public statement.

### 8.3 `euclid_proposition_26_saa_not_less`

This local helper is the formal core of the SAA proof. It packages the classical argument

$\text{strict side inequality} \longrightarrow \text{SAS} \longrightarrow \text{angle congruence} \longrightarrow \text{I.16 contradiction}.$

The helper is reusable under interchange of the two triangles, so the symmetric strict-inequality branch does not duplicate the proof.

### 8.4 `bookZero_31_trichotomy1`

This theorem converts the exclusion of both strict comparisons into segment congruence.

It is the precise formal replacement for the classical conclusion that, if neither segment is greater than the other, the two segments are equal.

## 9 Formalization Notes

Proposition I.26 is a particularly useful diagnostic theorem.

The ASA part shows what happens when a classical Euclidean proof has already been absorbed into the lower Hilbert interface. The Book I wrapper is then short and need not reproduce the historical reductio.

The SAA part shows the opposite situation. The available ASA theorem cannot be reached merely by permuting the vertices, because the given side is not the side between the two supplied angles. A genuine argument is still required.

The formal proof therefore mirrors an important part of Euclid’s historical structure: it constructs an interior point, applies SAS, and invokes the exterior-angle theorem I.16.

At the same time, the proof is reorganized to expose the reusable logical unit. Instead of proving two symmetric strict-inequality branches independently, one helper is applied twice.

This gives the architecture

classical geometric contradiction  $\longrightarrow$  named helper  $\longrightarrow$  segment trichotomy  $\longrightarrow$  existing ASA interface.

No new geometric axiom, numerical length, coordinate system, or numerical angle measure is introduced.

## 10 Dependency Summary

At the classical level:

```

ASA:
I.3-style segment cut
  |
I.4 (SAS)
  |
proper-angle contradiction
  |
equal corresponding side
  |
I.4
  |
triangle congruence

```

```

SAA:
segment cut
  |
I.4 (SAS)
  |
I.16
  |
contradiction
  |
equal corresponding side
  |
I.4
  |
triangle congruence

```

At the formal level:

```

ASA:
hilbert_asa_sides
  |

```

```

      v
two missing sides
      |
      v
HilbertSSS
      |
      v
euclid_proposition_26_asa

SAA:
SAS + Proposition I.16
      |
      v
euclid_proposition_26_saa_not_less
      |
      +-- exclude EF < BC
      |
      +-- exclude BC < EF
      |
      v
bookZero_31_trichotomy1
      |
      v
BC ~= EF
      |
      v
euclid_proposition_26_asa
      |
      v
euclid_proposition_26_saa

```

Thus Proposition I.26 exhibits both modes of the current reconstruction in a single theorem: one half collapses to established interface geometry, while the other still requires a substantial Euclidean argument.

## 11 Conclusion

Proposition I.26 completes the elementary triangle-congruence theory of the first part of Book I by establishing the ASA and SAA criteria.

The formal reconstruction makes the distinction between the two cases sharper than the classical presentation.

ASA is already essentially available in the Hilbert interface. The Book I proof only has to orient the triangles correctly, recover the missing sides, and use SSS for the remaining angle.

SAA cannot be reduced by permutation alone. Its proof requires the exterior-angle theorem I.16 to eliminate a hypothetical strict inequality of the corresponding sides. Once both strict inequalities are excluded, Book Zero segment trichotomy reduces SAA to ASA.

The proposition therefore gives a compact example of the layered architecture of the project:

Hilbert interface  $\longrightarrow$  Book Zero order theory  $\longrightarrow$  Euclidean congruence theorem.